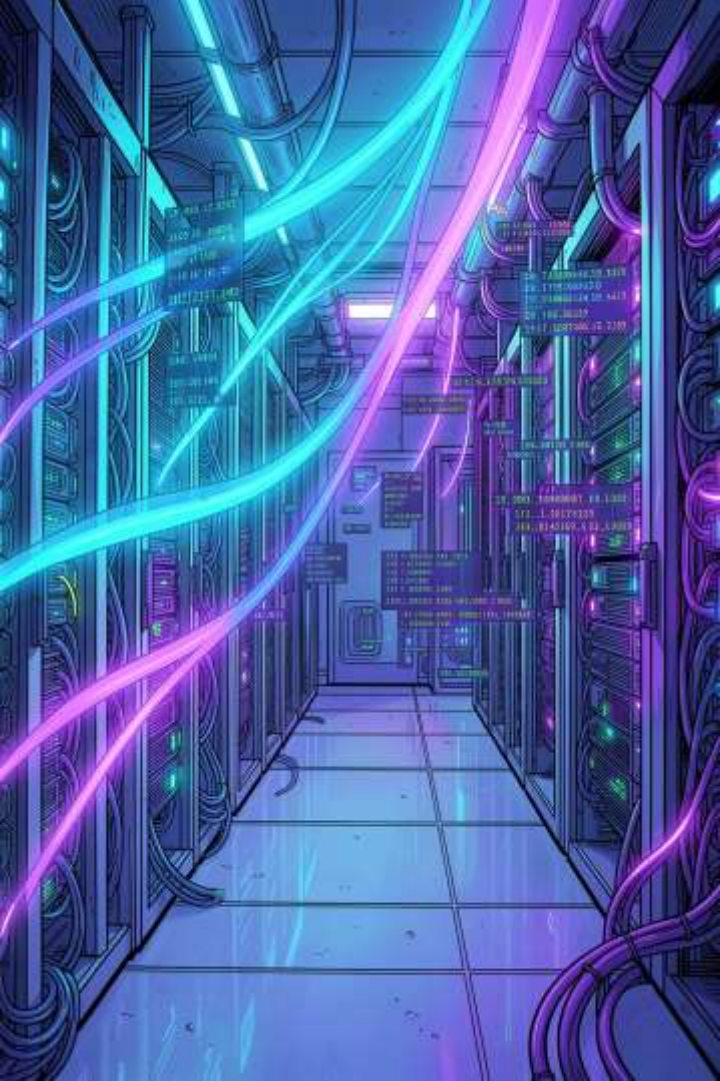




NEXUS

Log Anomaly Explainer

An AI-powered web application that transforms raw, complex system logs into clear, human-readable root-cause analysis — powered by Google Gemini AI.

AI INTEGRATION

FULL-STACK PROTOTYPE

DEVOPS TOOLING

The Problem

Logs Are Hard to Read



Application and server logs are dense, technical, and time-consuming to parse. Engineers spend valuable minutes — sometimes hours — manually scanning thousands of lines to find the root cause of a failure.

The picture can't be displayed.

No Context

Raw logs lack explanation of why an error occurred.

The picture can't be displayed.

No Guidance

There's no suggested fix — just cryptic error codes.

The picture can't be displayed.

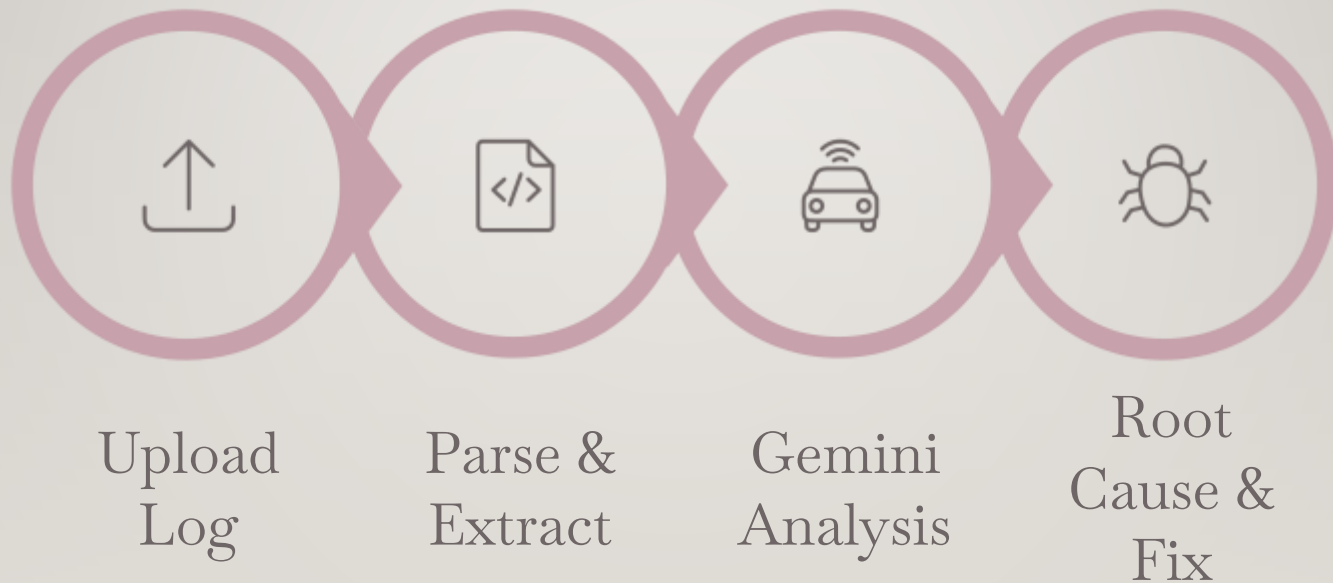
No Prioritization

Critical errors blend in with routine INFO messages.

Solution Overview

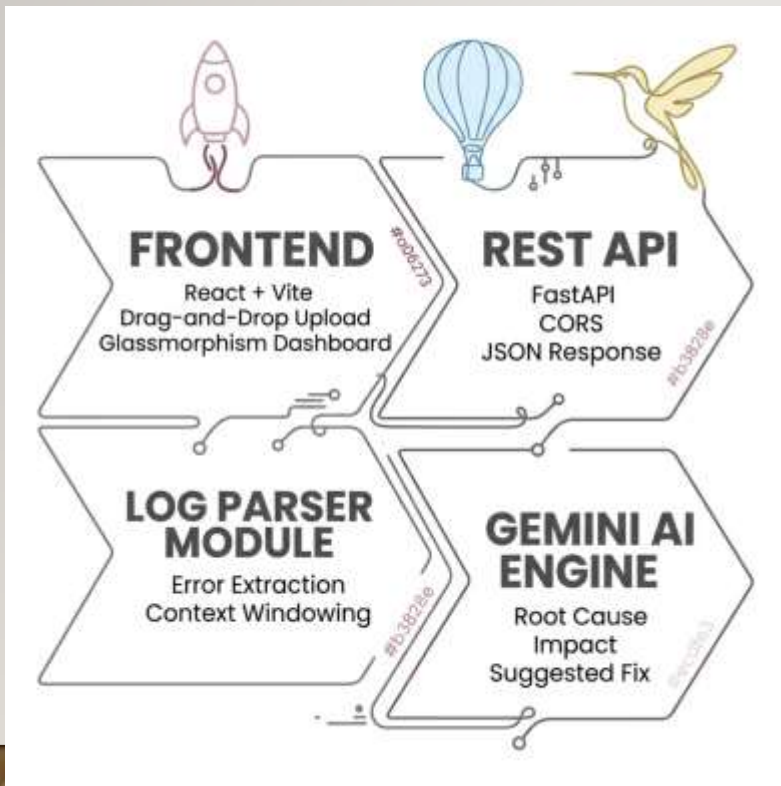
How It Works

The user uploads a `.log` file through a modern React frontend. The FastAPI backend parses the file, extracts error-relevant context, and sends it to Google's Gemini Flash model. Within seconds, the system returns a structured analysis — root cause, impact assessment, and a suggested fix — displayed in a clean, glassmorphic dashboard.



Zero database, zero authentication overhead — analysis happens in real time during a single session.

System Architecture



Design Principles

- **Stateless** — no database; files processed at runtime only
- **Modular** — separate `parser.py` and `agent.py` for clarity
- **Lightweight** — minimal dependencies, fast startup via Uvicorn
- **AI-Native** — Gemini Flash handles all semantic log interpretation



Environment variables (`.env` + `python-dotenv`) securely manage the Gemini API key.

Technology Stack

Built With Modern Tools



React + Vite

Fast, component-driven frontend with hot module replacement and a responsive glassmorphism UI.



Google Gemini Flash

Fast, cost-effective multimodal AI model used for structured log analysis and natural language explanation.



FastAPI + Uvicorn

Async Python backend with automatic OpenAPI docs, CORS support, and lightweight request handling.



Fetch API + REST

Clean HTTP communication between frontend and backend via POST requests with JSON responses.

Core Modules

What Each Component Does

File Upload

Drag-and-drop .log file ingestion with client-side validation.

Log Parser

Extracts error lines and surrounding context windows for focused AI input.

AI Analysis

Prompts Gemini (acting as a senior DevOps engineer) to generate Root Cause, Impact, and Fix.

Severity Detector

Classifies logs as **NORMAL** or **CRITICAL** based on keyword matching.

Result Dashboard

Renders severity badge, line count, AI explanation, and remediation steps.

Prompt Engineering for Consistent Output

The System Prompt

Gemini is instructed to behave as a **senior DevOps engineer**, ensuring every response follows a strict three-part structure:



01

Root Cause

What specifically caused this error?

02

Impact

What systems or users are affected?

03

Suggested Fix

Actionable remediation steps for the engineer.

Strengths & Trade-offs

Advantages & Limitations

✓ Advantages

- Simple, lightweight architecture — completable in hours
- No database required; stateless by design
- Fast AI-powered analysis via Gemini Flash
- Supports both application and server logs
- Beginner-friendly — great for learning AI integration

⚠️ Current Limitations

- Depends on Gemini API availability and quota
- Text-based .log files only
- No historical storage or log history
- No authentication or access control
- No real-time log streaming or monitoring

Roadmap

Future Enhancements

Phase 1 — Persistence — 1

Add MongoDB or PostgreSQL for log history and user sessions.

2

Phase 2 — Security

Implement user authentication, API rate limiting, and secure file handling.

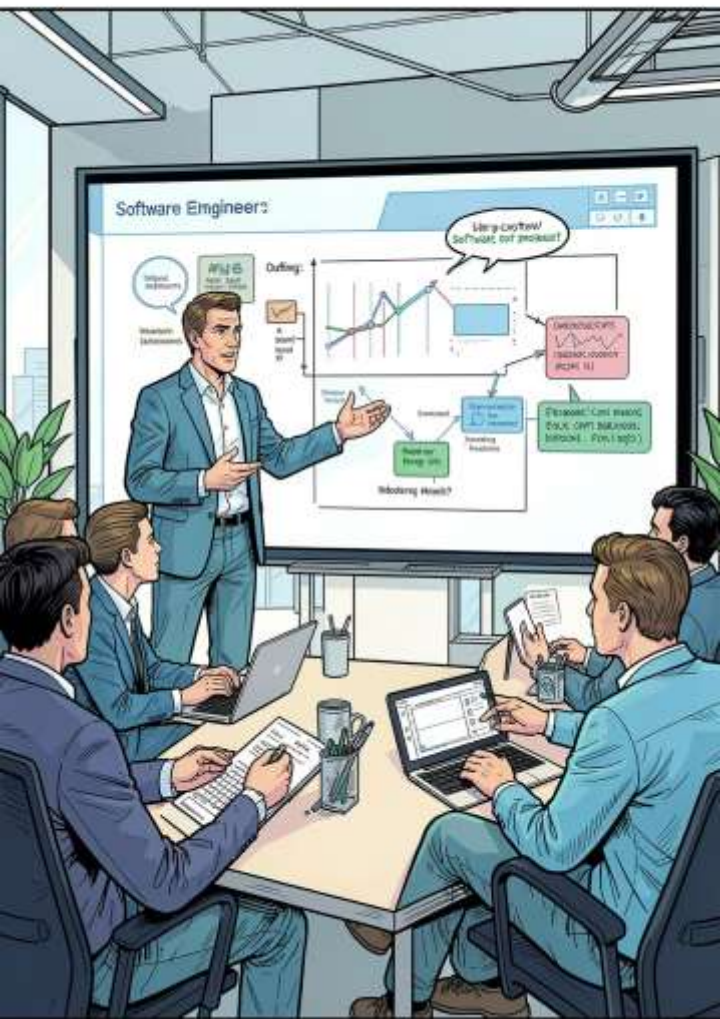
Phase 3 — Analytics — 3

Integrate Chart.js or Recharts for severity charts and anomaly trends.

4

Phase 4 — Scale

Real-time log streaming, multi-file batch analysis, email alerts, and cloud deployment.



Interview Ready

What This Project Demonstrates



Full-Stack Dev

React frontend + FastAPI backend + REST communication



AI Integration

Google Gemini API with structured prompt engineering



Log Analysis

Error extraction, severity classification, context windowing



Best Practices

Environment variables, modular design, CORS, async I/O



Core takeaway: This prototype shows how AI can simplify DevOps workflows — converting raw machine logs into actionable, human-readable insights in seconds.



Conclusion

- 1) Our project demonstrates how Generative AI can simplify software troubleshooting.
- 2) Instead of manually reading thousands of log entries, users receive intelligent explanations within seconds

Thank you